Unit 3: Combinatorics, Probability, and Statistics

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Table of Contents I

Table of Contents II

Table of Contents III

Combinatorics

Combinatorics

Definition

Combinatorics is the branch of mathematics concerned with counting.

The fundamental counting principle states:

If there are m_k ways to make the k-th decision for $k = 1, 2, \ldots n$,

then there are a total of

$$\prod_{k=1}^{n} m_k = m_1 \cdot m_2 \cdot \ldots \cdot m_n$$

ways to make the n independent decisions.

Combinatorics Example

Example

Suppose that your bank pin consists of six characters, each of which can be any letter or numerical digit.

(a) How many possible personal codes are there?

(b) If you want your pin to have no repeated characters, how many possible pins would there be now?

Permutations and Combinations

Definition

If the order in which the objects are chosen matters, each complete selection of k objects from an original n is called a **permutation**. If the order does not matter, then each complete selection is called a **combination**.

Permutations and Combinations without Replacement

The number of *permutations* of k elements from a selection of n *without replacement* is

$$_nP_k = \frac{n!}{(n-k)!}.$$

The number of *combinations* of k elements from a selection of n *without replacement* is

$$_nC_k = \binom{n}{k} = \frac{n!}{k!(n-k)!}.$$

Permutations and Combinations without Replacement

Example

(a) Among eight comparable investment funds, *ex ante*, how many different ways are there to list the top three performing funds?

(b) How many three-element subsets does a set of nine elements have?

n Elements of k Types

Suppose there are n_k indistinguishable elements of type k, where $k = 1, 2, \ldots m$. Let

$$n = n_1 + n_2 + \ldots + n_m.$$

Then the number of *unique* rearrangements of the n elements is

$$\binom{n}{n_1, n_2, \ldots, n_m} = \frac{n!}{n_1! n_2! \ldots n_m!}.$$

n Elements of k Types

Example

How many unique rearrangements of the letters in the word BANANA are there?

Stars-and-Bars

Theorem (Stars-and-Bars)

Suppose there are k types of items. If the order in which items are selected does not matter, the number of ways to select n items is

$$\binom{n+k-1}{k-1} = \binom{n+k-1}{n}.$$

Stars-and-Bars Example

Example

In how many ways can we write the number six as the sum of four non-negative integers?

Stars-and-Bars on YouTube

Watch Ken Ribet explain Stars-and-Bars (`https://youtu.be/UTCScjoPymA`).

The Pigeonhole Principle

Theorem (Pigeonhole Principle)

Suppose there are n objects distributed among m sets, where $n \geq m$.

Then one set must contain at least $\lceil n/m \rceil$ elements.

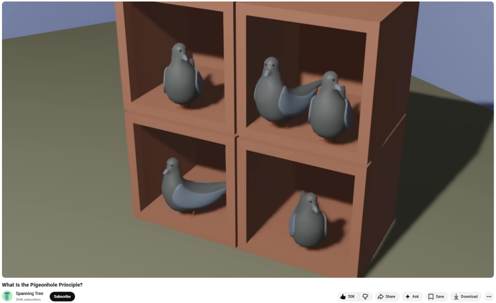
The Pigeonhole Principle Example

Example

Suppose we have a set of $n+1$ whole numbers. Show that we can always select two elements a and b such that $a-b$ is divisible by n.

The Pigeonhole Principle on YouTube

This video has some interesting applications of the pigeonhole principle (`https://youtu.be/B2A2pGrDG8I?si=q-HjtxMYPneSaICJ`).



Inclusion–Exclusion Principle

Theorem (Inclusion–Exclusion Principle)

Suppose $|A|$ denotes the number of elements in A. Consider finite sets A_1 and A_2. The number of elements in $A_1 \cup A_2$ is

$$|A_1 \cup A_2| = |A_1| + |A_2| - |A_1 \cap A_2|.$$

More generally, for finite sets $A_1, A_2, \ldots A_n$, the number of elements in $\bigcup_{i=1}^{n} A_i$ is

$$\left|\bigcup_{i=1}^{n} A_i\right| = \sum_{i=1}^{n} |A_i| - \sum_{1 \le i < j \le n} |A_i \cap A_j| + \sum_{1 \le i < j < k \le n} |A_i \cap A_j \cap A_k|$$
$$- \ldots + (-1)^{n+1} \left|A_1 \cap A_2 \cap \ldots \cap A_n\right|.$$

Inclusion–Exclusion Principle Example

Example

There are 80 students enrolled in the UCLA MFE class of 2027. Forty-five students are taking statistical arbitrage, 43 are taking credit markets, and 37 are taking behavioral finance. If 23 students are taking statistical arbitrage and credit markets, 15 are taking statistical arbitrage and behavioral finance, and 20 are taking credit markets and behavioral finance, how many students are taking all three classes?

Probability

σ-Algebra Definition

Definition

Let Ω be some set, and let 2^Ω represent the set of all subsets of Ω. Then a subset $\mathcal{F}$ of 2^Ω is called a **σ-algebra** if and only if it satisfies the following three properties:

- SA1 The set Ω is in $\mathcal{F}$.
- SA2 If $A \in \mathcal{F}$, then $A^c = \{x \in \Omega \mid x \notin A\} \in \mathcal{F}$.
- SA3 If $A_k \in \mathcal{F}$ for $k = 1, 2, \ldots$, then $\displaystyle\bigcup_{k=1}^{\infty} A_k \in \mathcal{F}$.

Probability Definition

Definition

Let Ω be a set and let $\mathcal{F}$ be a σ-algebra of Ω. A **probability** is defined as a function $P : \mathcal{F} \to [0, 1]$ if the following hold:

P1 $P(\varnothing) = 0$.

P2 $P(\Omega) = 1$.

P3 For $E_k \in \mathcal{F}$ and $E_k \cap E_\ell = \varnothing$ for $k \neq \ell$, we have

$$P\left(\bigcup_{k=1}^{\infty} E_k\right) = \sum_{k=1}^{\infty} P(E_k).$$

We typically call Ω the **sample space**. The elements of $\mathcal{F}$ are called **events**.

Altogether, $(\Omega, \mathcal{F}, P)$ forms a **probability space**.

Probability Example

Example

Consider the sample space $[0,1] \times [0,1]$. We define $P : \mathcal{F} \to [0,1]$ such that $P(E) = \text{area}(E)$. In this case, we would have

$$\mathcal{F} = \{E \subseteq [0,1] \times [0,1] \mid \text{area of } E \text{ exists}\}.$$

We need $\mathcal{F}$ because there are subsets of $[0,1] \times [0,1]$ whose area cannot be measured!

Properties of Probability Measure

Consider the probability space $(\Omega, \mathcal{F}, P)$. Suppose A and B are in $\mathcal{F}$.

- $A \subseteq B$ implies $P(A) \leq P(B)$
- $P(A^c) = 1 - P(A)$
- $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

Probability Example

Example

The point (x, y) is selected at random from $[0, 1] \times [0, 1]$. Assume that all points within the sample space are equally likely to be selected. What is the probability that xy is less than $1/2$?

Conditional Probability Definition

Definition

If the occurrence of the event A depends on the occurrence of B then the conditional probability is

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

whenever $P(B) > 0$. If $P(B) = 0$ the conditional probability is undefined.

Conditional Probability Example

Credit	Excellent	Good	Poor	Σ
Defaults	930	1,350	5,856	8,136
Non-Defaults	10,321	18,007	13,536	41,864
Σ	11,251	19,357	19,392	50,000

Example

A loan originator has a 50,000 observation loan data set. This originator places applicants' credit into one of three categories: excellent, good, and poor. Details about this data set are shown above. What is the probability of default given that an applicant's credit is placed into the category good?

Bayes' Rule

Theorem (Bayes)

Suppose $P(B) > 0$. *Then*

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}.$$

Theorem (Law of Total Probabilities)

Suppose $B_i \cap B_j = \varnothing$ *for* $i \neq j$ *and* $\bigcup_{i=1}^{n} B_i = \Omega$. *Then*

$$P(A) = P(A|B_1)P(B_1) + P(A|B_2)P(B_2) + \ldots + P(A|B_n)P(B_n).$$

Bayes' Rule Example

Example

Assume the probability that an unskilled asset manager beats the market is 50%, and the probability that a skilled asset manager beats the market is 60%. If 10% of asset managers are skilled, what is the probability that an asset manager is skilled given that she beats the market?

Bayes' Rule on YouTube

Watch the StatQuest video about Bayes' theorem (a.k.a. Bayes' rule) on YouTube (`https://youtu.be/9wCnvr7Xw4E`).

Independent Events

Definition

Events A and B are independent if $P(A|B) = P(A)$.

If two events are independent then it is easy to prove

$$P(A \cap B) = P(A)P(B).$$

Independent Events Example

Example

One urn contains four red balls and six blue balls. A second urn contains sixteen red balls and b blue balls. A single ball is drawn from each urn. The probability that both balls are the same color is 0.44 . Calculate b.

Random Variable Definition

Definition

A random variable X is a function $X : \Omega \to \mathbb{R}$ where $P\left(X^{-1}(A)\right)$ exists for every $A \subseteq \mathbb{R}$.

The probability that X takes on values in the subset S of $\mathbb{R}$ is written

$$P(X \in S) = P\left(\{\omega \in \Omega \mid X(\omega) \in S\}\right).$$

The probability that X takes on value x in $\mathbb{R}$ is written

$$P(X = x) = P\left(\{\omega \in \Omega \mid X(\omega) = x\}\right).$$

Types of Random Variables

There are three types of random variables: *discrete*, *continuous*, and *mixed*.

- *Discrete:* Range is finite or countably infinite (distinct, separate values).
- *Continuous:* Range is uncountable (values form a continuous interval or union of intervals).
- *Mixed:* Exhibits characteristics of both discrete (probabilities at specific points) and continuous (probabilities over intervals).

Examples of Types of Random Variables

Example

State whether the random variables are discrete, continuous, or mixed.

(a) A coin is tossed ten times. The random variable X is the number of tails.

(b) The random variable Y measures the time until a firm defaults on its debt.

(c) The random variable Z measures the time, in years, until a firm defaults on a particular bond that matures in T years. If the firm meets all of its payment obligations for the bond, Z is given a value of T.

Probability Mass Function Definition

Definition

For a discrete random variable X, we define the **probability mass function** or **pmf** to be

$$p(x) = P(X = x).$$

Probability Mass Function Example

Example

A coin is flipped two times. Let the random variable X denote the number of heads. Find the pmf of X.

Cumulative Distribution Function Definition

Definition

For a discrete random variable X, we define the **cumulative distribution function** or **cdf** to be

$$F(x) = P(X \leq x) = \sum_{t \leq x} P(X = t).$$

Cumulative Distribution Function Example

Example

The random variable X denotes the number selected from the set $\{1, 2, 3\}$. If each number is equally likely, find the cdf.

Discrete Uniform Distribution

Definition

The **discrete uniform distribution** is a symmetric probability distribution.

Each of the n values has equal probability $1/n$ of being observed.

Binomial Distribution

Definition

The **binomial distribution** with parameters n and p is the discrete probability distribution of the number of successes in a sequence of n independent experiments, each ending in either success with probability p or failure with probability $q = 1 - p$.

To denote that X is a binomial random variable with parameters n and p, we write $X \sim \mathcal{B}(n, p)$. The probability

$$P(X = x) = \begin{cases} \binom{n}{x} p^x q^{n-x}, & x = 0, 1, \ldots, n \\ 0, & \text{otherwise.} \end{cases}$$

Binomial Distribution Example

Example

Suppose a portfolio manager has a portfolio of 25 high yield bonds. She assumes that each bond has a one-year default probability of 5%. Her portfolio is well diversified so defaults are independent. What is the probability that more than one bond defaults?

Binomial Distribution Python Example

Example

Use Python to check the previous result by simulating the event 100,000 times.

Solution.

```
# Import modules
import numpy as np
from scipy.stats import binom

# Generate 100,000 simulations
defaults = binom.rvs(n = 25, p = 0.05, loc = 0, size = 100_000, random_state = 0)

# Calculate probability
prob = np.mean(defaults > 1)

print(f'The probability that more than one bond defaults is {prob:.3f}.')
```

The output is 0.357.

Poisson Distribution

Definition

The **Poisson distribution** is a discrete probability distribution that expresses the probability of a given number of events x occurring in a fixed interval if these events occur independently with a constant rate λ.

To denote that X is a Poisson random variable with parameter λ, we write $X \sim \mathcal{P}(\lambda)$. The probability

$$P(X = x) = \begin{cases} \dfrac{\lambda^x e^{-\lambda}}{x!}, & x = 0, 1, 2, \ldots \\ 0, & \text{otherwise.} \end{cases}$$

Poisson Distribution Example

Example

After graduation from the MFE program, you become a famous and successful portfolio manager. To help younger men and women succeed in finance, you begin writing a memoir of your life. In the first draft of your book, there are an average of three spelling errors per page. Suppose that the number of errors per page follows a Poisson distribution. What is the probability of having no errors on a page?

Continuous Random Variable Definition

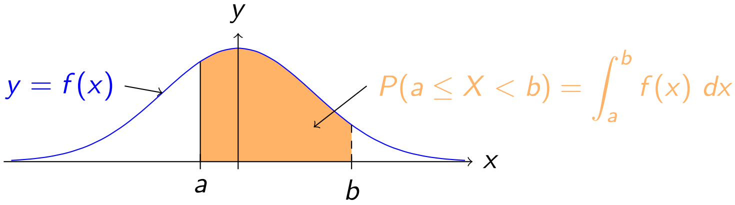
Definition

We say that a random variable X is **continuous** if there exists a non-negative function f defined for all real numbers having the property that for any set B of real numbers we have

$$P(X \in B) = \int_B f(x)\, dx.$$

The function f is called the **probability density function** or **pdf** of the random variable X.

Probability Density Function Plot



Cumulative Distribution Function Definition

Definition

The **cumulative distribution function** or **cdf** of a continuous random variable X is defined to be

$$F(x) = P(X \leq x) = \int_{-\infty}^{x} f(t)\, dt,$$

where f is the pdf of X.

Properties of CDFs

The cumulative distribution function F of a continuous random variable X with probability density function f satisfies the following properties.

(a) $0 \leq F(x) \leq 1$

(b) If $x_1 \leq x_2$, then $F(x_1) \leq F(x_2)$

(c) $\lim_{x \to -\infty} F(x) = 0$

(d) $\lim_{x \to \infty} F(x) = 1$

(e) $P(a < X \leq b) = F(b) - F(a)$

(f) F is continuous

(g) $F'(x) = f(x)$ whenever F' exists

All of these, except (f) and (g), hold for discrete random variables as well.

PDF and CDF Example

Example

Suppose a random variable X's pdf is proportional to x^2 on the interval $[-1, 2]$ and 0 elsewhere. Find the pdf and cdf of X.

The Continuous Uniform Distribution Definition

Definition

The **continuous uniform distribution** models a random variable where every value within a specific interval is equally likely to occur. The distribution is defined by two parameters: a (the minimum value) and b (the maximum value). It is commonly denoted as $\mathcal{U}(a, b)$.

The pdf and cdf of the continuous uniform distribution on $[a, b]$ are

$$f(x) = \begin{cases} \dfrac{1}{b-a}, & a \le x \le b \\ 0, & \text{otherwise} \end{cases} \quad \text{and} \quad F(x) = \begin{cases} 0, & x < a \\ \dfrac{x-a}{b-a}, & a \le x \le b \\ 1, & x > b. \end{cases}$$

Continuous Uniform Distribution Example

Example

Suppose that $X \sim \mathcal{U}(0, b)$, where $b > 1$. Find $P(X^2 < X)$.

Normal Distribution Definition

Definition

The **normal** or **Gaußian distribution** has probability density function

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}.$$

The parameters of the distribution are μ and σ^2. To denote that X follows a normal distribution, we write $X \sim \mathcal{N}(\mu, \sigma^2)$.

Normal Distribution Python Example

Example

Use Python to illustrate how μ and σ affect the graph of the pdf.

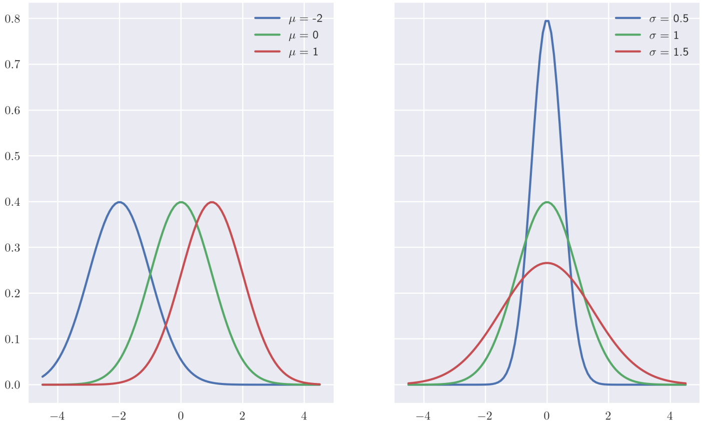
Solution.

```
# Import modules
import numpy as np, matplotlib.pyplot as plt
from scipy.stats import norm

# Use LaTeX and increase resolution
plt.rcParams['text.usetex'] = True
plt.rcParams['figure.dpi'] = 300

# Use Seaborn style
plt.style.use('seaborn-v0_8')

# Mu- and sigma-values
mu_vals, sigma_vals = [-2, 0, 1], [0.5, 1, 1.5]

# Set up subplots
fig, ax = plt.subplots(1, 2, sharey = True, figsize = (10, 6))

# Get x-values
x_vals = np.linspace(-4.5, 4.5, 100)
```

Normal Distribution Python Example

```
# Loop over mu-values
for mu in mu_vals:
    # Note norm.pdf vectorized
    y_vals = norm.pdf(x_vals, loc = mu, scale = 1)

    ax[0].plot(x_vals, y_vals, label = fr'$\mu =$ {mu}')

# Loop over sigma-values
for sigma in sigma_vals:
    # Note norm.pdf vectorized; also scale is std not var
    y_vals = norm.pdf(x_vals, loc = 0, scale = sigma)

    ax[1].plot(x_vals, y_vals, label = fr'$\sigma =$ {sigma}')

# Create legend for subplots
ax[0].legend()
ax[1].legend()

# Save the figure
plt.savefig(path + r'ex3-1.png')

# Show plot
plt.show()
```

Normal Distribution Python Example Result



Standard Normal Distribution Definition

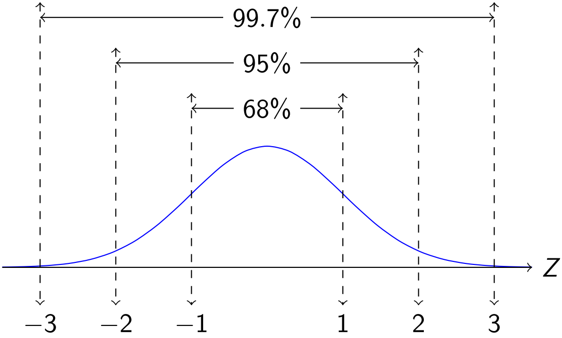
Definition

The **standard normal distribution** is $\mathcal{N}(0, 1^2)$, i.e. it is a normal distribution with $\mu = 0$ and $\sigma^2 = 1$. The pdf and cdf are sometimes denoted ϕ and Φ, respectively.

You can translate a normal random variable into a standard normal variable by calculating its Z-score:

$$X \sim \mathcal{N}(\mu, \sigma^2) \qquad \text{implies} \qquad Z = \frac{X - \mu}{\sigma} \sim \mathcal{N}(0, 1^2).$$

The Empirical Rule



If $Z \sim \mathcal{N}(0, 1^2)$, then

$$P(|Z| < 1) \approx 68\%, \qquad P(|Z| < 2) \approx 95\%, \qquad \text{and} \qquad P(|Z| < 3) = 99.7\%.$$

Normal Distribution Example

Example

Suppose that the daily returns for a stock index $R \sim \mathcal{N}(0, 0.014^2)$. What is the probability of this index dropping more than 4.2%?

Exponential Distribution Definition

Definition

The **exponential distribution** models the time elapsed between events in a process where events occur continuously and independently at a constant average rate λ.

To denote that the random variable X follows an exponential distribution, we write $X \sim \mathcal{E}(\lambda)$. The pdf and cdf of an exponential distribution are

$$f(x) = \begin{cases} 0, & x < 0 \\ \lambda e^{-\lambda x}, & x \geq 0 \end{cases} \quad \text{and} \quad F(x) = \begin{cases} 0, & x < 0 \\ 1 - e^{-\lambda x}, & x \geq 0. \end{cases}$$

Exponential random variables are often used to model arrival times, waiting times, and defaults.

Exponential Distribution Example

Example

Suppose that the time until default, measured in years, of a firm T follows an exponential distribution with $\lambda = 0.10$. What is the probability that $2 < T < 5$?

Expected Value Definition

Definition

The **expected value** of a discrete random variable X is

$$E[X] = \mu_X = \sum_x xP(x),$$

where we sum over the range of X. If X is a continuous random variable, then the expected value is

$$E[X] = \mu_X = \int_{-\infty}^{\infty} xf(x)\, dx.$$

Expected Value Example 1

Example

Prove that if $X \sim \mathcal{E}(\lambda)$, then $E[X] = \dfrac{1}{\lambda}$.

Expected Value Example 2

Example

Suppose that the pmf of N is

$$p(n) = \begin{cases} \left(\frac{1}{2}\right)^n, & n = 1, 2, \ldots \\ 0, & \text{otherwise.} \end{cases}$$

Calculate $E[N]$.

Linearity of Expected Values

For random variables X and Y, and scalars α and β, we have

$$E[\alpha X + \beta Y] = \alpha E[X] + \beta E[Y].$$

Variance and Standard Deviation Definitions

Definition

The **variance** of a discrete random variable X is

$$\text{Var}(X) = \sigma_X^2 = \sum_x (x - \mu_X)^2 \, p(x) = E[(x - \mu_X)^2].$$

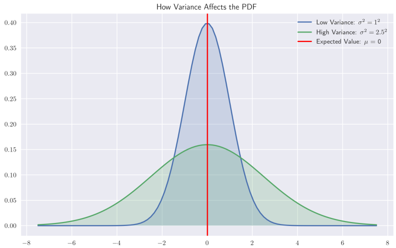
If X is a continuous random variable

$$\text{Var}(X) = \sigma_X^2 = \int_{-\infty}^{\infty} (x - \mu_X)^2 \, f(x) \; dx = E[(x - \mu_X)^2].$$

The **standard deviation** of X is the square root of its variance and is denoted by σ_X.

What is Variance?

Variance represents the spread of data points around their expected value. It measures how far the random variable tends to differ from the expected value. Low variance indicates the outputs of the random variable are clustered together, while high variance shows that outputs are more spread out.



Variance Example

Example

Prove the identity

$$\mathrm{Var}(X) = E[X^2] - (E[X])^2 \,.$$

Property of Variance

For the random variable X and scalars α and β

$$\mathrm{Var}\left(\alpha X + \beta\right) = \alpha^2 \mathrm{Var}(X).$$

Expected Value and Variance Formulas

Distribution	Expected Value	Variance
Discrete Uniform	$\frac{1}{n}\sum_{k=1}^{n} x_k$	$\frac{1}{n}\sum_{k=1}^{n} x_k^2 - \left(\frac{1}{n}\sum_{k=1}^{n} x_k\right)^2$
$\mathcal{B}(n,p)$	np	$np(1-p)$
$\mathcal{P}(\lambda)$	λ	λ
$\mathcal{U}(a,b)$	$\frac{b+a}{2}$	$\frac{(b-a)^2}{12}$
$\mathcal{N}(\mu,\sigma^2)$	μ	σ^2
$\mathcal{E}(\lambda)$	$\frac{1}{\lambda}$	$\frac{1}{\lambda^2}$

k-th Moment Definitions

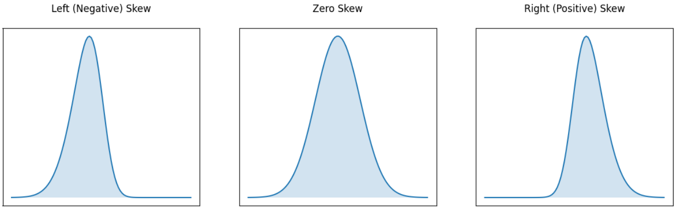
Definition

- If X is a random variable, then its ***k*-th moment** is $E[X^k]$.
- If $\mu = E[X]$ and $\sigma^2 = \text{Var}(X)$, then the **standardized *k*-th moment** is $E\left[\left(\frac{X-\mu}{\sigma}\right)^k\right]$.

Skew Definition

Definition

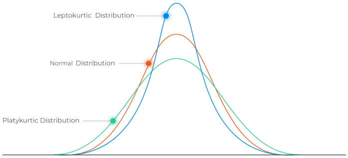
The standardized third moment is the **skew**.



Kurtosis Definition

Definition

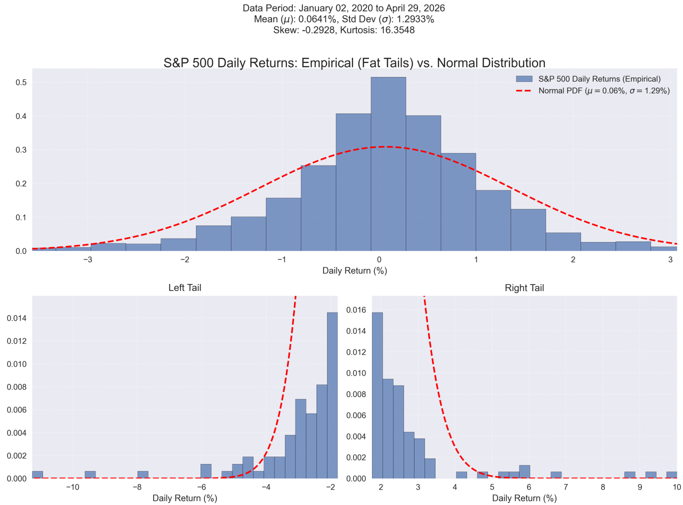
The standardized fourth moment is the **kurtosis**. A normal distribution has kurtosis of 3. If the kurtosis is greater than 3, it is **leptokurtic**, and if the kurtosis is less than 3 it **platykurtic**.



The kurtosis is typically considered a measure of tail thickness, though the true the meaning is more nuanced:

https://stats.stackexchange.com/questions/172467/what-is-the-meaning-of-tail-of-kurtosis.

The S&P 500, Fat Tails, Skewness, and Kurtosis



Moment Generating Function

Definition

The **moment generating function** of a random variable X is

$$\psi(t) = E[e^{tX}].$$

We call ψ the moment generating function of X because

$$E[X^k] = \psi^{(k)}(0).$$

To see why this is the case, it is helpful to note

$$e^{tX} = 1 + tX + \frac{(tX)^2}{2!} + \frac{(tX)^3}{3!} + \dots.$$

Moment Generating Function Example

Example

Find the moment generating function of X, where X is the discrete uniform distribution with support $\{1, 2, 3\}$.

Function of a RV Python Example

Example

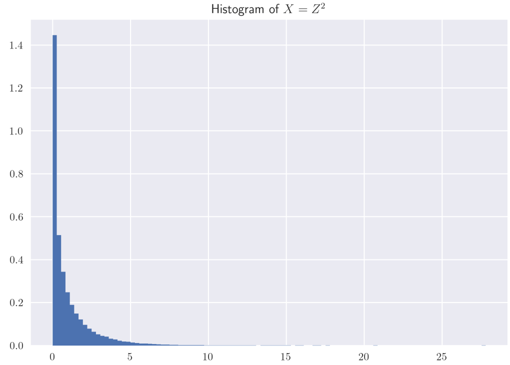
Suppose $Z \sim \mathcal{N}(0,1)$. Consider $X = Z^2$. Use Python to approximate the graph of the pdf of X using a histogram.

Solution.

```
# Import modules
import numpy as np, matplotlib.pyplot as plt
from scipy.stats import norm

# Use LaTeX and increase image resolution
plt.rcParams['text.usetex'], plt.rcParams['figure.dpi'] = True, 300

# Use Seaborn style
plt.style.use('seaborn-v0_8')

# Generate Z
Z = norm.rvs(size = 100_000)

# Calculate X
X = Z**2

# Generate histogram; make sure density is True
plt.hist(X, bins = 100, density = True)

# Get title
plt.title(r'Histogram of $X = Z^2$')

# Save the figure
plt.savefig(path + r'ex3-3.png')

plt.show()
```

Function of a Random Variable Python Result

This is a known distribution $\chi^2(1)$. See Wikipedia for more details.



Function of a Random Variable

The pdf can be obtained analytically by considering the cdf and differentiating.

Function of a Random Variable Example

Example

Suppose $Z \sim \mathcal{N}(0, 1)$. Consider $X = Z^2$. Calculate the pdf of X.

Function of a Random Variable Theorem

Theorem

Let X be a random variable with pdf f. Suppose $P(x_1 < x < x_2) = 1$. Let $Y = r(X)$ and suppose r is differentiable and one-to-one for $x_1 < x < x_2$. Let the image of (x_1, x_2) under r be (y_1, y_2). If s is the inverse of r, then the pdf of Y is

$$g(y) = \begin{cases} f\left(s(y)\right)\left|\frac{ds}{dy}\right|, & y_1 < y < y_2 \\ 0, & \text{otherwise.} \end{cases}$$

Function of a Random Variable Example

Example

Suppose $X \sim \mathcal{U}(0, 1)$ and let $Y = \frac{1}{27}X^3$. Find the pdf of Y.

Joint Continuous Definition

Definition

Two random variables X and Y are said to be **jointly continuous** if there exists a function $f_{XY}(x, y) \geq 0$ with the property that for every subset C of $\mathbb{R}^2$, we have

$$P\Big((X, Y) \in C\Big) = \iint_C f_{XY}(x, y)\; dA.$$

The function $f_{XY}(x, y)$ is called the **joint probability density function** of X and Y, and the **joint cumulative distribution function** is

$$F(x, y) = \int_{-\infty}^{y} \int_{-\infty}^{x} f(s, t)\; ds\, dt.$$

Joint Distributions Example

Example

Suppose

$$f_{XY}(x,y) = \begin{cases} \frac{1}{2}, & -1 \leq x \leq y \leq 1 \\ 0, & \text{otherwise.} \end{cases}$$

Then $P(2X - Y > 0) =$

Marginal PDF

The marginal pdfs of X and Y given joint pdf f_{XY} are

$$f_X(x) = \int_{-\infty}^{\infty} f(x,y)\ dy \qquad \text{and} \qquad f_Y(y) = \int_{-\infty}^{\infty} f(x,y)\ dx.$$

Independent Events

If the random variables X and Y are independent, then

$$f_{XY}(x, y) = f_X(x)f_Y(y).$$

Independent Events Example

Example

Suppose X and Y represent the normalized daily trading volume (scaled between 0 and 1) for two correlated tech equities. Their joint probability density function is given by

$$f_{XY}(x,y) = \begin{cases} x+y, & 0 \le x \le 1, \quad 0 \le y \le 1 \\ 0, & \text{otherwise.} \end{cases}$$

Are the trading volumes X and Y independent?

Change of Variables

Theorem

Let X and Y be continuous random variables with joint density f_{XY}. Assume that there is a subset S of $\mathbb{R}^2$ such that $P\Big((X,Y) \in S\Big) = 1$. Let

$$U = r_1(X,Y) \qquad \text{and} \qquad V = r_2(X,Y),$$

where r_1 and r_2 define a one-to-one differentiable transformations from S onto some subset T of $\mathbb{R}^2$. If

$$X = s_1(U,V) \qquad \text{and} \qquad Y = s_2(U,V)$$

describe the inverse transformation, where s_1 and s_2 are also one-to-one and differentiable. Then the joint pdf of U and V is

$$g_{UV}(u,v) = \begin{cases} f_{XY}\Big(s_1(u,v), s_2(u,v)\Big)\,|det(J)|\,, & (u,v) \in T \\ 0, & \text{otherwise.} \end{cases}$$

where

$$J = \begin{pmatrix} \frac{\partial s_1}{\partial u} & \frac{\partial s_1}{\partial v} \\ \frac{\partial s_2}{\partial u} & \frac{\partial s_2}{\partial v} \end{pmatrix}$$

is the Jacobian.

Change of Variables Example

Example

Suppose $f_{XY}(x, y) = \frac{1}{2\pi} e^{-\frac{x^2+y^2}{2}}$. Find the joint pdf of r and θ, where $x = r\cos\theta$, $y = r\sin\theta$, $r \geq 0$, and $0 \leq \theta < 2\pi$.

Covariance Definition

Definition

Let X and Y be random variables having finite means. Suppose $E[X] = \mu_X$ and $E[Y] = \mu_Y$. The **covariance of X and Y** is

$$\mathrm{Cov}(X, Y) = E\Big[(X - \mu_X)(Y - \mu_Y)\Big].$$

A useful identity is

$$\mathrm{Cov}(X, Y) = E[XY] - E[X]E[Y].$$

Covariance Example

Example

Suppose $X \sim \mathcal{U}(-1, 1)$. Calculate the covariance between X and X^2.

Properties of Covariance

Suppose X_i and Y_j are random variables for all i and j, and α is a real number.

(a) $\text{Cov}(X_1, X_2) = \text{Cov}(X_2, X_1)$

(b) $\text{Cov}(X_1, X_1) = \text{Var}(X_1)$

(c) $\text{Cov}(\alpha X_1, X_2) = \alpha \text{Cov}(X_1, X_2)$

(d) $\text{Cov}(X, Y) = 0$ if X and Y are independent

(e) $\text{Cov}\left(\sum_{i=1}^{m} X_i, \sum_{j=1}^{n} Y_j\right) = \sum_{i=1}^{m}\sum_{j=1}^{n} \text{Cov}(X_i, Y_j)$

Correlation Definition

Definition

Let X and Y be random variables with finite variances σ_X^2 and σ_Y^2, respectively. Then the **correlation of X and Y** is

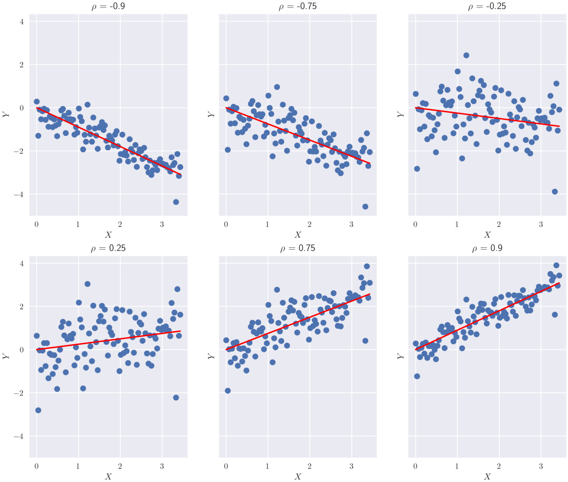
$$\rho(X, Y) = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y}.$$

In Python the correlation between two variables can be calculated using `stats.pearsonr` within `scipy`. See the SciPy documentation for more details.

Correlation Plot Python Code

```python
# Import modules
import numpy as np
import matplotlib.pyplot as plt
from scipy.stats import norm

# Use LaTeX
plt.rcParams['text.usetex'] = True

# Increase resolution
plt.rcParams['figure.dpi'] = 300

# Use Seaborn style
plt.style.use('seaborn-v0_8')

# Generate standard normals
Z = norm.rvs(size = 100)

# List correlations
rho_vals = [-0.9, -0.75, -0.25, 0.25, 0.75,
        0.9]

# Set up subplots
fig, ax = plt.subplots(2, 3, sharey = True,
        figsize = (12, 10))

for i, rho in enumerate(rho_vals):

    # Get X
    X = np.linspace(0, 1, len(Z))

    # Make sure variance of X is 1
    X /= np.std(X)

    # Get Y
    Y = rho * X + np.sqrt(1 - rho**2) * Z

    # Get row and column
    row, col = i//3, i%3

    # Draw scatter plot
    ax[row, col].scatter(X, Y)

    # Draw plot of underlying relationship
    ax[row, col].plot(X, rho * X, color = '
     red')

    # Add title to subplot
    ax[row, col].set_title(rf'$\rho$ = {rho
     }')

    # Add labels for x and y axes
    ax[row, col].set_xlabel(r'$X$')
    ax[row, col].set_ylabel(r'$Y$')

# Save the figure
plt.savefig(path + r'ex3-4.png')

plt.show()
```

Correlation Plot



Variance of Sum

Theorem

If X and Y are random variables and α and β are constants, then

$$Var(\alpha X + \beta Y) = \alpha^2 Var(X) + \beta^2 Var(Y) + 2\alpha\beta Cov(X, Y).$$

Variance of Sum Example

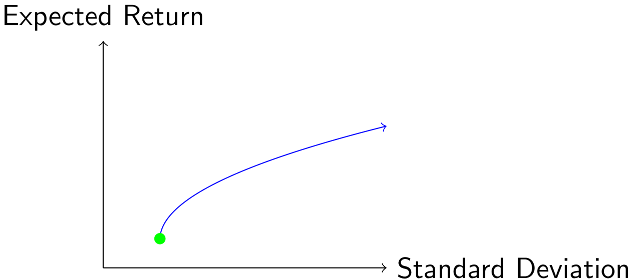
Example

Suppose $X \sim \mathcal{U}(-1, 1)$. Calculate the variance of $X - 2X^2$.

Efficient Frontier Definition

Definition

The **efficient frontier** is the set of portfolios which satisfy the condition that no other portfolio in the investment universe exists with a higher expected return and the same standard deviation of return.



Portfolio Construction Example

Example

Consider assets 1 and 2 with monthly expected returns of 1% and 0.5% and standard deviations of 10% and 6%, respectively. Suppose the correlation between returns is 25%.

(a) Find the minimum variance portfolio.

(b) Graph the efficient frontier.

Assume that you cannot take short positions in this investment universe.

Portfolio Construction Example Python Code

```python
# Import modules
import numpy as np, pandas as pd, matplotlib.pyplot as plt
from scipy.optimize import minimize_scalar
from matplotlib.ticker import PercentFormatter

# Use LaTeX and increase resolution
plt.rcParams['text.usetex'] = True
plt.rcParams['figure.dpi'] = 300

# Use Seaborn style
plt.style.use('seaborn-v0_8')

# Define expected returns
mu1, mu2 = 0.01, 0.005

# Define standard deviations
sigma1, sigma2 = 0.10, 0.06

# Define correlation
rho = 0.25

# Create function for portfolio expected return
port_mu = lambda w: mu1 * w + mu2 * (1 - w)

# Create function for portfolio standard deviations
def port_sigma(w):

    # Calculate variance...

    # ... uncorrelated variance
    var = w**2 * sigma1**2 + (1 - w)**2 * sigma2**2

    # ... variance due to correlation
    var += 2 * w * (1 - w) * rho * sigma1 * sigma2

    # Take square root to obtain standard deviation
    return np.sqrt(var)
```

Portfolio Construction Example Solution

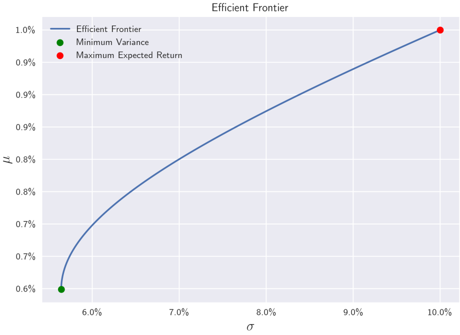
```python
# Get weights for minimum volatility portfolio; no short positions
# Since the square root is monotonic for positive inputs same as
# minimum variance portfolio
w_min_var = minimize_scalar(port_sigma, bounds = (0, 1), method = 'bounded').x

# Define number of samples
samples = 100

# Get weights
wt_vals = np.linspace(w_min_var, 1.0, samples)

# Calculate expected returns for each wt
mu_vals = port_mu(wt_vals)

# Calculate std of returns for each wt
sigma_vals = port_sigma(wt_vals)

# Stack results, so dimension (-1, 2)
port_results = np.column_stack([mu_vals, sigma_vals])

# Convert to data frame
port_results = pd.DataFrame(port_results, columns = ['mu', 'sigma'])
```

Portfolio Construction Example Solution

```python
# Plot efficient frontier
plt.plot(port_results['sigma'], port_results['mu'], label = 'Efficient Frontier')

# Add dot for minimum variance portfolio
plt.scatter(port_results.loc[0, 'sigma'], port_results.loc[0, 'mu'],
            label = 'Minimum Variance', color = 'green', zorder = 3)

# Add dot for maximum expected return
plt.scatter(port_results.loc[samples - 1, 'sigma'], port_results.loc[samples - 1, 'mu'],
            label = 'Maximum Expected Return', color = 'red', zorder = 3)

# Add legend
plt.legend()

# Add x-label
plt.xlabel(r'$\sigma$', fontsize = 15)

# Add y-label
plt.ylabel(r'$\mu$', fontsize = 15)

# Add title
plt.title('Efficient Frontier')

# Write both axes as a percentage
plt.gca().xaxis.set_major_formatter(PercentFormatter(1.0, decimals = 1))
plt.gca().yaxis.set_major_formatter(PercentFormatter(1.0, decimals = 1))

# Save the figure
plt.savefig(path + r'ex3-5.png')

plt.show()

print(r'The minimum variance portfolio has respective weights in ',
      rf'assets 1 and 2 of {w_min_var:.1%} and {1 - w_min_var:.1%}.',
     f'The standard deviation of the minimum variance portfolio is {port_sigma(w_min_var)
     :.1%}.')
```

Portfolio Construction Result

The minimum variance portfolio has respective weights in assets 1 and 2 of 19.8% and 80.2%. The standard deviation of the minimum variance portfolio is 5.6%.



Statistics

Markov Inequity Theorem

Theorem (Markov Inequity)

Suppose X is a random variable such that $P(X \geq 0) = 1$. Then for each real number $t > 0$,

$$P(X \geq t) \leq \frac{E[X]}{t}.$$

Chebyshev Inequality

Theorem (Chebyshev Inequality)

Let X be a random variable for which $Var(X)$ exists. Then for every number $t > 0$,

$$P\Big(|X - E[X]| \geq t\Big) \leq \frac{Var(X)}{t^2}.$$

Mean and Variance of Sample Means

Theorem

Let $X_1, X_2, \ldots, X_n$ be an independent random sample from a distribution with mean μ and variance σ^2. Let $\bar{X}_n$ be the sample mean. Then $E[\bar{X}_n] = \mu$ and $Var(\bar{X}_n) = \sigma^2/n$.

Chebyshev Inequality Example

Example

Suppose $E[X] = 0$ and $\text{Var}(X) = 1$. Use the Chebyshev Inequality to find a minimum independent sample size n, so that $P\left(|\bar{X}_n| \geq 0.5\right) \leq 0.01$?

Convergence in Probability Definition

Definition

A sequence $X_1, X_2, \ldots$ of random variables **converges in probability** to X if for each $\epsilon > 0$,

$$\lim_{n \to \infty} P\Big(|X_n - X| < \epsilon\Big) = 1.$$

We often write $X_n \xrightarrow{p} X$ to denote that the sequence converges in probability to X.

Law of Large Numbers

Theorem (Law of Large Numbers)

Suppose that $X_1, X_2, \ldots, X_n$ form a random sample from a distribution with mean μ and finite variance. Let $\bar{X}_n$ denote the sample mean. Then

$$\bar{X}_n \xrightarrow{p} \mu$$

as $n \to \infty$.

Law of Large Numbers Python Example

Example

Use an exponential distribution with $\lambda = 1$ to illustrate the Law of Large Numbers. Generate 10 paths of cumulative averages and show they converge to $1/\lambda = 1$.

Law of Large Numbers Python Example Solution

```python
# Import modules
import numpy as np, matplotlib.pyplot as plt, time
from scipy.stats import expon

# Use LaTeX and increase resolution
plt.rcParams['text.usetex'], plt.rcParams['figure.dpi'] = True, 300

# Use Seaborn style
plt.style.use('seaborn-v0_8')

# Set random seed
np.random.seed(0)

# Define lambda
lam = 1

# Generate exponentials; mu is scale = 1/lambda for exponential
X = expon.rvs(scale = 1/lam, size = (100_000, 10))

# Define n-values
n_vals = np.arange(1, X.shape[0] + 1)

# Calculate cumulative sample means
X = X.cumsum(axis = 0)
X = X/n_vals[:, np.newaxis]

# Define the minimum n on the plots
min_n_val = 100
# Only interested in observations after the first 100
X = X[min_n_val - 1:, :]
n_vals = n_vals[min_n_val - 1:]
```

Law of Large Numbers Python Example Solution

```
# Plot results
plt.plot(n_vals, X, linewidth = 1)

# Add dashed line for expected value
plt.axhline(y = 1/lam, linewidth = 1, linestyle = 'dashed', color = 'red',
  label = r'$\mu$')

# Add label to x-axis
plt.xlabel('$n$')

# Add label to y-axis
plt.ylabel(r'$\bar{X}_n$')

# Add legend
plt.legend()

# Give plot a title
plt.suptitle(f'Law of Large Numbers Using Exponential Distribution\n Each of the {X.shape
     [1]} paths converges to the expected value of {1/lam}')

# Add a subtitle
plt.title(r'$\bar{X}_n\stackrel{p}{\longrightarrow} \mu$ as $n\to \infty$')

# Change the horizontal axis limits
plt.xlim(min_n_val, n_vals.max())

# Save the figure
plt.savefig(path + r'ex3-6.png')

plt.show()

# Clear RAM
del X
```

Law of Large Numbers Python Example Result

Convergence in Distribution Definition

Definition

A sequence $X_1, X_2, \ldots$ of random variables with cdfs $F_1, F_2, \ldots$ **converge in distribution** to the random variable X with cdf F if

$$\lim_{n \to \infty} F_n(x) = F(x)$$

for every number x at which F is continuous.

We often write $X_n \xrightarrow{d} X$ to denote convergence in distribution.

Empirical CDF (ECDF) Definition

Definition

Suppose we have observed data $x_1, x_2, \ldots, x_n$ which are sampled from the same distribution. The empirical cdf (or ecdf) given these data is

$$F_n(x) = \frac{1}{n}\sum_{k=1}^{n} H(x - x_k),$$

where the Heaviside function

$$H(x) = \begin{cases} 1, & x \geq 0 \\ 0, & \text{otherwise.} \end{cases}$$

Empirical CDFs in Python

In Python, it is very easy to code the empirical cdf:

```
ecdf = lambda x, sample:  np.mean(sample <= x).
```

You can also use the `ecdf` function in `scipy.stats`. See the Scipy documentation for more details.

Empirical CDF Python Example

Example

Graph the empirical cdf of a $\chi^2(1)$ distribution with samples of sizes 5, 50, and 1000 as well as the true cdf of the $\chi^2(1)$ to show that the discrete uniform distribution on sampled $\chi^2(1)$ values converges in distribution to the $\chi^2(1)$ distribution.

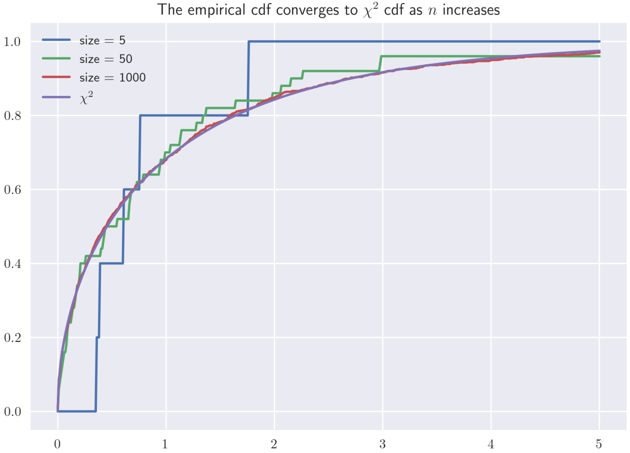
Empirical CDF Python Example Solution

```python
# Import modules
import numpy as np, matplotlib.pyplot as plt
from scipy.stats import chi2

# Use LaTeX and increase resolution
plt.rcParams['text.usetex'], plt.rcParams['figure.dpi'] = True, 300

# Use Seaborn style
plt.style.use('seaborn-v0_8')

# Set random seed
np.random.seed(0)

# Create empirical cdf
ecdf = lambda x, sample: np.mean(sample <= x)

# List of sample sizes
sample_sizes = [5, 50, 1000]

# Create x-values for plot
x_vals = np.linspace(0, 5, 500)
```

Empirical CDF Python Example Solution

```python
# Loop over the sample sizes
for sample_size in sample_sizes:

    # Generate sample
    sample = chi2.rvs(df = 1, size = sample_size)

    # Get the y-values
    y_vals = [ecdf(x, sample) for x in x_vals]

    # Plot values
    plt.plot(x_vals, y_vals, label = f'size = {sample_size}')

# Get y-values; function vectorized
y_vals = chi2.cdf(x_vals, df = 1)

# Plot chi^2 cdf
plt.plot(x_vals, y_vals, label = r'$\chi^2$')

# Clear up RAM
del sample, x_vals, y_vals

# Show legend
plt.legend()

# Add title
plt.title(r'The empirical cdf converges to $\chi^2$ cdf as $n$ increases')

# Save the figure
plt.savefig(path + r'ex3-7.png')

plt.show()
```

Empirical CDFs Python Example Result



Central Limit Theorem

Theorem (Central Limit Theorem)

If the random variables $X_1, X_2, \ldots, X_n$ are a random independent sample of size n from a given distribution with mean μ and finite variance σ^2, then

$$\frac{\sqrt{n}}{\sigma}\left(\bar{X}_n - \mu\right) \xrightarrow{d} \mathcal{N}(0, 1)$$

as $n \to \infty$.

In other words, the sample mean should approximately follow the distribution $\mathcal{N}\left(\mu, \frac{\sigma^2}{n}\right)$ when n is large.

Central Limit Theorem Python Example

Example

Illustrate the Central Limit Theorem with an exponential distribution, using the empirical cdf and means of samples of sizes 2, 5, and 100.

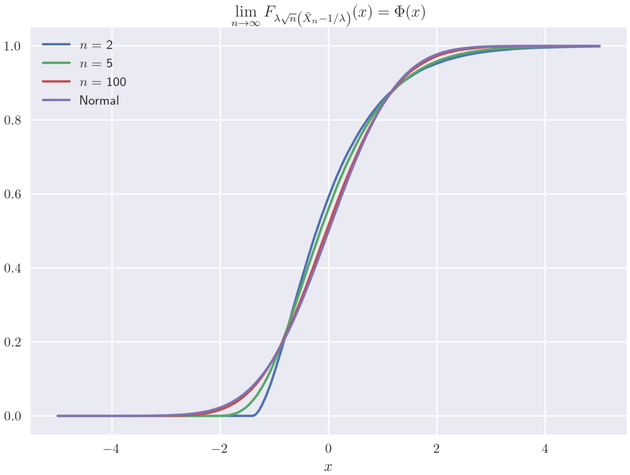
Central Limit Theorem Python Example Solution

```
# Import modules
import numpy as np
import matplotlib.pyplot as plt
from scipy.stats import expon, norm

# Use LaTeX
plt.rcParams['text.usetex'] = True

# Increase resolution
plt.rcParams['figure.dpi'] = 300

# Use Seaborn style
plt.style.use('seaborn-v0_8')

# Set random seed
np.random.seed(0)

# Create empirical cdf
ecdf = lambda x, sample: np.mean(sample <= x)

# Create x-values for plot
x_vals = np.linspace(-5, 5, 500)

# Create n-values
n_vals = [2, 5, 100]

# Let's do 100,000 trials
trials = 100_000
```

Central Limit Theorem Python Example Solution

```
# Scale is 1/lambda
lam = 1
# Loop over the n-values
for n in n_vals:
    # Generate the sample; mean is 1/lam and std is 1/lam
    sample =  expon.rvs(scale = 1/lam, size = (n, trials)).mean(axis = 0)

    # Change sample so it has mean 0 and sd 1
    sample = np.sqrt(n)/(1/lam) * (sample - 1/lam)

    # Get y-values
    y_vals = [ecdf(x, sample) for x in x_vals]

    # Plot values
    plt.plot(x_vals, y_vals, label = f'$n =$ {n}')

# Get y-values
y_vals = norm.cdf(x_vals)

# Plot standard normal cdf
plt.plot(x_vals, y_vals, label = 'Normal')

# Add x-axis label
plt.xlabel('$x$')

# Show legend
plt.legend()

# Add title
plt.title(r'$\displaystyle\lim_{n\to\infty}F_{\lambda\sqrt{n}\left(\bar{X}_n - 1/\lambda\
    right)}(x) = \Phi(x)$')

# Save the figure
plt.savefig(path + r'ex3-8.png')

plt.show()
```

Central Limit Theorem Python Example Result



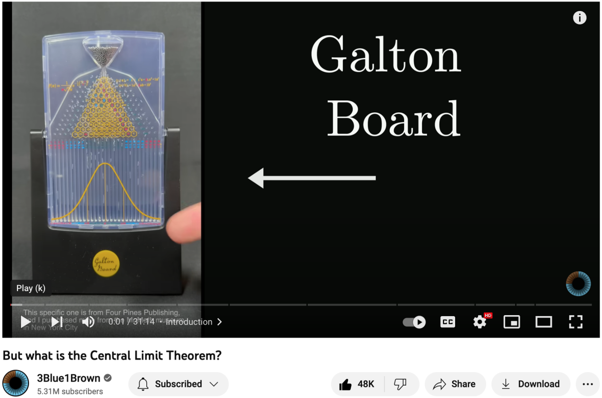
Central Limit Theorem Example

Example

A regional bank issues a portfolio of 100 independent small business loans. Based on historical data, the expected loss on each individual loan due to default is $16,000, with a standard deviation of $3,000. The risk management desk has allocated a maximum capital reserve of $1,660,000 to cover losses for this entire portfolio. What is the probability that the total portfolio loss exceeds the capital reserve?

Central Limit Theorem on YouTube

3Blue1Brown has a great video on the Central Limit Theorem (https://youtu.be/zeJD6dqJ5lo)



Hypothesis Testing Example

Example

You sample the numbers 8, 0.5, 0, and -0.5 from a normal distribution with known variance of 1. Perform a hypothesis test to determine whether the mean of the distribution is statistically different from 0 at a significance level of 5%.

Hypothesis Testing Steps

1. State the hypotheses.
2. Set the criteria for a decision.
3. Compute the test statistic.
4. Make a decision.

State the Hypotheses Definition

Definition

- The **null hypothesis** (H_0) is a statement about a population parameter that is assumed to be true. We will test the plausibility of the observed data given the assumption of a true null hypothesis.
- An **alternative hypothesis** (H_1) is a statement that directly contradicts the null hypothesis by claiming that the actual value of a population parameter is less than, greater than, or not equal to the value stated in the null hypothesis.

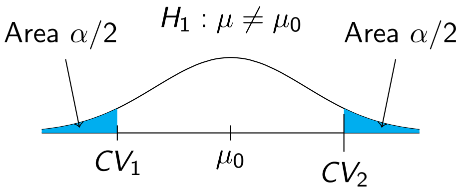
Set the Criteria for a Decision

Definition

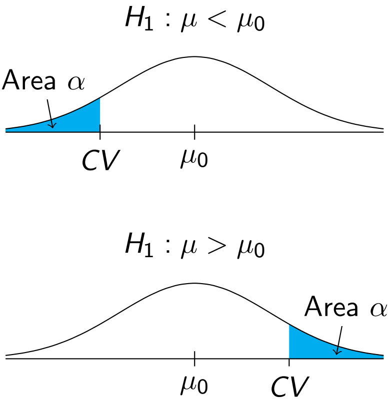
The **significance level** (α) is a threshold probability that denotes how strong the evidence must be to reject the null hypothesis. If the probability of observing the data given that the null hypothesis is true is less than α, we conclude that the observed data is not consistent with the null hypothesis and deem the result statistically significant.

Hypothesis Test for $H_0 : \mu = \mu_0$

Two Tailed Test



One Tailed Tests



Compute the Test Statistic

Definition

A **test statistic** is a scalar quantity derived from the sample data that summarizes the evidence against the null hypothesis (H_0). The probability distribution of this statistic under H_0 is known, allowing us to quantify the extremity of the observed data.

Make a Decision

Definition

The ***p*-value** is the probability of an observation being at least as extreme as the test statistic assuming the null hypothesis is true.

- If the *p*-value is greater than or equal to α, we *fail to reject the null hypothesis*.
- If the *p*-value is less than α, we *reject the null hypothesis*.

Types of Errors

		Decision	
		Fail to Reject Null	Reject Null
Truth	H_0 True	Correct $1-\alpha$	Type I Error α
	H_0 False	Type II Error β	Correct $1-\beta$

Confidence and Power Definitions

Definition

- The **confidence level** of a hypothesis test is $1 - \alpha$.
- The **power** of a hypothesis test is $1 - \beta$.

Types of Hypothesis Tests

Incomplete list of types of hypothesis tests.

Name	Test Statistic	Comments
One-sample z-test	$z = \frac{\bar{x}-\mu_0}{\sigma/\sqrt{n}}$	Normal population or large n and known σ. The value μ_0 is the mean assuming the null hypothesis.
One-sample t-test	$t = \frac{\bar{x}-\mu_0}{s/\sqrt{n}}$	Normal population or large n and unknown σ. The degrees of freedom are $df = n - 1$
One-proportion z-test	$z = \frac{\hat{p}-p_0}{\sqrt{p_0(1-p_0)/n}}$	p_0 is the proportion under the null hypothesis and $\min\{n \cdot p_0, n \cdot (1 - p_0)\} > 10$

Population Parameters vs. Sample Estimators

	Population	**Sample**
Mean	$\mu = E[X]$	$\bar{x} = \frac{1}{n}\sum_{k=1}^{n} x_k$
Variance	$\sigma^2 = E[(X-\mu)^2]$	$s^2 = \frac{1}{n-1}\sum_{k=1}^{n}(x_k - \bar{x})^2$
Standard Deviation	$\sigma = \sqrt{E[(X-\mu)^2]}$	$s = \sqrt{\frac{1}{n-1}\sum_{k=1}^{n}(x_k - \bar{x})^2}$

Sample Standard Deviation

$$s = \sqrt{\frac{1}{n-1}\sum_{k=1}^{n}(x_k - \bar{x})^2}$$

Sample Standard Deviation Python Example

Example

Find the distribution of the biased and sample standard deviations when ten elements are sampled from $\mathcal{N}(0, 1^2)$.

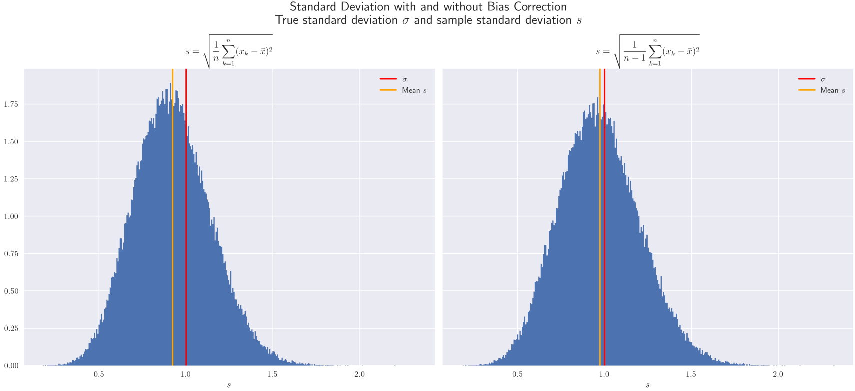
Sample Standard Deviation Python Example Solution

```
# Import modules
import numpy as np, matplotlib.pyplot as plt
from scipy.stats import norm

# Use LaTeX and increase resolution
plt.rcParams['text.usetex'], plt.rcParams['figure.dpi'] = True, 300

# Use Seaborn style
plt.style.use('seaborn-v0_8')

# Set random seed
np.random.seed(0)

# Define number of trials
trials = 100_000

# Generate normal rvs with mean 0 and variance 1
Z = norm.rvs(loc = 0, scale = 1, size = (10, trials))

# Calculate the sd without bias correction
sd0 = np.apply_along_axis(np.std, 0, Z, ddof = 0)

# Calculate the sd with bias correction
sd1 = np.apply_along_axis(np.std, 0, Z, ddof = 1)

# Define number of bins
num_bins = int(np.sqrt(trials))
```

Sample Standard Deviation Python Example Solution

```
# Set up subplots
fig, ax = plt.subplots(1, 2, sharex = True, sharey = True, figsize = (15, 7))

ax[0].hist(sd0, bins = num_bins, density = True)
ax[0].axvline(x = 1, color = 'red', label = r'$\sigma$')
ax[0].axvline(x = np.mean(sd0), color = 'orange', label = r'Mean $s$')

ax[0].legend()
ax[0].title.set_text(
  r'$s = \displaystyle\sqrt{\frac{1}{n}\sum_{k = 1}^n (x_k - \bar{x})^2}$')
ax[0].set_xlabel(r'$s$')

ax[1].hist(sd1, bins = num_bins, density = True)
ax[1].axvline(x = 1, color = 'red', label = r'$\sigma$')
ax[1].axvline(x = np.mean(sd1), color = 'orange', label = r'Mean $s$')

ax[1].legend()
ax[1].title.set_text(
  r'$s = \displaystyle\sqrt{\frac{1}{n - 1}\sum_{k = 1}^n (x_k - \bar{x})^2}$')
ax[1].set_xlabel(r'$s$')

# Give entire figure title
fig.suptitle(r'Standard Deviation with and without Bias Correction' + '\n'
  + r'True standard deviation $\sigma$ and sample standard deviation $s$',
             fontsize = 16)

# Add padding
fig.tight_layout(pad = 1)

# Save the figure
plt.savefig(path + r'ex3-9.png')

plt.show()
```

Sample Standard Deviation Python Example Result



Unbiased Estimation of Standard Deviation

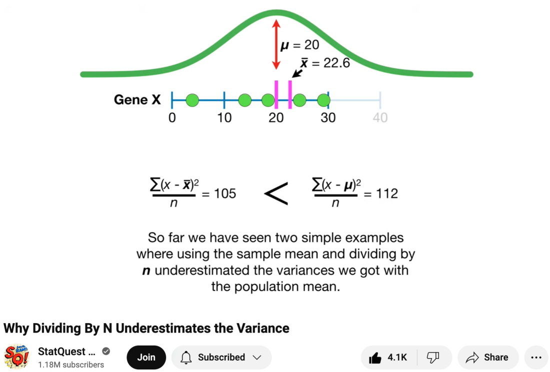
In fact,

$$\frac{1}{n-1}\sum_{k=1}^{n}(x_k-\bar{x})^2$$

is an unbiased estimate of $\text{Var}(X)$, but the correction does not servive the square root transformation. See the Wikipedia article Unbiased estimation of standard deviation for more details.

Sample Variance on YouTube

There's a StatQuest about sample variance (`https://www.youtube.com/watch?v=sHRBg6BhKjI`)!



Student's t-Distribution

Definition

Let $X_1, X_2, \ldots X_n \sim \mathcal{N}(\mu, \sigma^2)$ be independent and identically distributed samples. Then

$$T = \frac{\bar{X} - \mu}{s/\sqrt{n}}$$

follows a t-distribution with $n-1$ degrees of freedom. We often write $T \sim t_{n-1}$.

For $n > 30$ the t-distribution is approximately equal to a normal distribution. As a result, this distribution is only relevant for small samples.

Hypothesis Testing with Unknown Variance

Example

You sample the numbers 8, 0.5, 0, and -0.5 from a normal distribution. Perform a hypothesis test to determine whether the mean of the distribution is larger than 0 at a significance level of 10%. Note: If $T \sim t_3$, then $P(T > 1.638) \approx 10\%$.

Hypothesis Testing Proportions

Example

You flip a coin 100 times and it lands heads 65 times. Perform a hypothesis test to determine whether the probability of landing heads is statistically different from 50% at a 95% confidence level.

Criticism of p-Values

The use of p-values has been criticized for several reasons. Some of them include:

- **Misinterpretation of Probability:** A p-value is not the probability that the null hypothesis is true. It measures the compatibility of the data with a specific model, not the likelihood of the hypothesis itself.
- **Relationship to Chance:** It is not the probability that effects were produced by "random chance." It is computed *assuming* the null model is true.
- **Arbitrary Thresholds:** The 5% and 10% levels are purely conventional. There is no qualitative scientific difference between a p-value of 4.9% and 5.1%. See the American Statistical Association (ASA)'s statement.
- **Effect Size vs. Significance:** A p-value does not indicate the size or importance of an effect. Large sample sizes can produce "significant" p-values for trivial effects.
- **Assumption-Dependent:** Calculating p-values may require numerous assumptions about the data. Interpreting them may require subjective judgment regarding the plausibility of those assumptions.
- **Multiple Testing & p-hacking:** p-values often fail to control for multiple testing. This is a common issue in financial backtesting, where researchers may rework results until they achieve significance.

Confidence Interval

The confidence interval contains the population parameter with probability $1 - \alpha$.

- Known Variance:

$$\left(\bar{x} + z_{\alpha/2} \frac{\sigma}{\sqrt{n}}, \bar{x} + z_{1-\alpha/2} \frac{\sigma}{\sqrt{n}} \right).$$

- Unknown Variance:

$$\left(\bar{x} + t_{\alpha/2} \frac{s}{\sqrt{n}}, \bar{x} + t_{1-\alpha/2} \frac{s}{\sqrt{n}} \right).$$

  Degrees of freedom are $n - 1$.

- Proportion:

$$\left(\hat{p} + z_{\alpha/2} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}, \hat{p} + z_{1-\alpha/2} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}} \right).$$

Confidence Interval Python Example

Example

Repeatedly sample 10 elements from a standard normal distribution and calculate the confidence interval for the mean to illustrate a confidence interval at a significance level of 10% with unknown variance.

Confidence Interval Python Example Solution

```python
# Import modules
import numpy as np
import matplotlib.pyplot as plt
from scipy.stats import norm, t

# Use LaTeX and increase resolution
plt.rcParams['text.usetex'], plt.rcParams['figure.dpi'] = False, 300

# Use Seaborn style
plt.style.use('seaborn-v0_8')

# Set random seed
np.random.seed(0)

# True values are mu = 0 and sigma = 1
mu, sigma = 0, 1

# Samples of size 10
sample_size = 10

# Set significance level
alpha = 0.10

# Calculate lower and upper t-values
t_lower = t.ppf(alpha/2, df = sample_size - 1)
t_upper = t.ppf(1 - alpha/2, df = sample_size - 1)

# How many trials?
trials = 100

# Create counter of times mu is in the interval
num_mu_in_interval = 0
```

Confidence Interval Python Example Solution

```python
for i in range(trials):

    # Generate sample
    sample = norm.rvs(loc = mu, scale = sigma, size = sample_size)

    # Calculate x_bar and s
    x_bar, s = np.mean(sample), np.std(sample, ddof = 1)

    # Calculate theoretical confidence interval
    lower_bound = x_bar + s/np.sqrt(sample_size) * t_lower
    upper_bound = x_bar + s/np.sqrt(sample_size) * t_upper

    # Check if mu is in the interval
    in_interval = lower_bound < mu < upper_bound

    # Plot sample points
    plt.scatter(sample, sample_size * [i], color = 'gray', s = 5,
                alpha = 0.5)

    # Define color depending on if interval contains mu
    color = 'blue' if in_interval else 'red'

    # Plot confidence interval
    plt.plot([lower_bound, upper_bound], [i, i], color = color)

    # Add to count
    num_mu_in_interval += int(in_interval)
```

Confidence Interval Python Example Solution

```
# Add a vertical line where mu is
plt.axvline(x = mu, color = 'black', label = r'$\mu$')

# Don't need the vertical integer numbering
plt.yticks([])

# Add a title
plt.title('Confidence Interval')

# Add annotation for results
plt.annotate(f'Experimental: {num_mu_in_interval/trials:0.0%}',
             xy = (1.8, trials))
plt.annotate(f'Theoretical: {1 - alpha:0.0%}%',
             xy = (1.8, trials - 5))

# Add legend
plt.legend()

# Save the figure
plt.savefig(path + r'ex3-10.png')

plt.show()
```

Confidence Interval Python Example Result